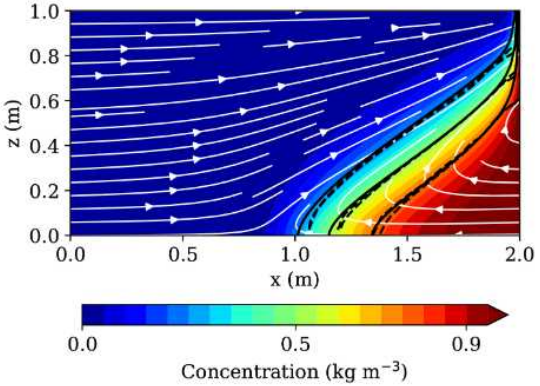
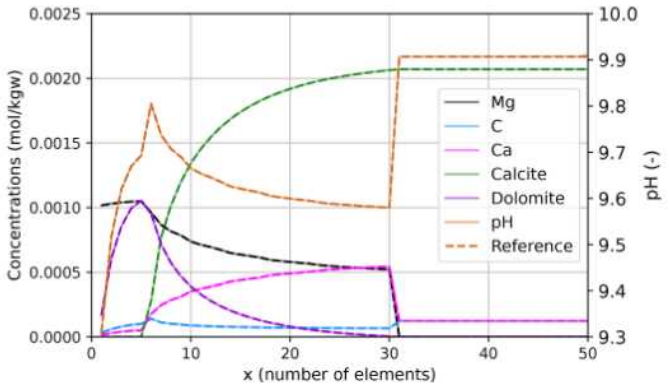
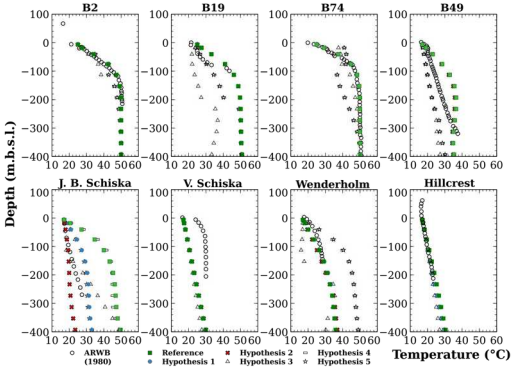
TRANSPORT Simulation Environment

Introduction

Thomas Kempka
GFZ German Research Centre for Geosciences
University of Potsdam







Why Yet Another Numerical Simulator (YANS), not even providing the features of existing ones?

- We have extensively tested {put any transport simulator here} and are not satisfied with the capabilities of any of these
- We do not want to spend multiple months to understand simulators other people have developed (we have been doing this over the past 14 years :-/)
- We want to be flexible and couple/integrate processes as required (T, H, C, M?!, R, B, E)
- We want to understand what we are doing and especially any potential limitations, so that we do NOT leave our scientific work to black boxes
- We have decided that we would like to abandon complex grids, but rather focus on processes and their coupling (→ FDM suffices)
- Most of us are really tired of developing workarounds to get coupled process simulations working, producing questionable results
- We need a simple tool that WE are able to understand, use and adapt

TRANSPORTSE (YANS) benefits in a nutshell...

- FDM implementation of coupled density-driven Darcy flow, heat and species transport < 1500 lines(!) of Python code
- Explicit, semi-implicit and fully implicit solution of coupled equations (simple iterative SOR Gauss-Seidel solver)
- Just-in-time (JIT) compilation (numba) provides execution times in the same order of C/C++/FORTRAN codes
- EOS implementation is straight forward (parallelisation feasible)
- Chemical module coupling straight forward (parallelisation feasible)
- Runs in parallel on multiple CPU cores already scales with 2 cores on your lousy ;-) laptop (tested with 80 cores on GFZ cluster)
- Very efficient and simple model parametrisation thanks to numpy
- Generally very good Python code readability supports understanding of TRANSPORTSE implementation and eases modifications
- HOWEVER, crap in → crap out: you are the modeller and responsible for your work

TRANSPORTSE solves coupled systems of equations for density-driven Darcy flow, species and heat transport

Density-driven Darcy flow (Pressure formulation)

$$\rho_f\left((1-\phi)\alpha+\phi\beta\right)\frac{\partial P}{\partial t}=\nabla\left[\frac{k\rho_f}{\mu_f}\left(\nabla P-\rho_f g\nabla z\right)\right]+W$$

Diffusive and advective species transport (one PDE system for each species)

$$\phi\frac{\partial C}{\partial t}=\nabla(\phi D\nabla C-qC)+Q$$

Conductive and convective heat transfer

$$\left((1-\phi)c_r\rho_r+\phi c_f\rho_f\right)\frac{\partial T}{\partial t}=\nabla\left[\left((1-\phi)\lambda_r+\phi\lambda_f\right)\nabla T-qc_f\rho_f T\right]+H$$

Non-linearity due to parameter dependency on pressure, temperature and dissolved species/single-phase fluid composition and chemical reactions (R)

$$\rho_f(P,T,C),\beta(P,T,C),\mu_f(P,T,C),\phi(R),k(\phi),c_r(T),c_f(P,T,C),\lambda_r(T),\lambda_f(P,T,C)$$

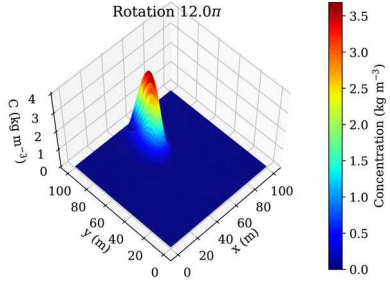
Generic model verification based on density-driven flow

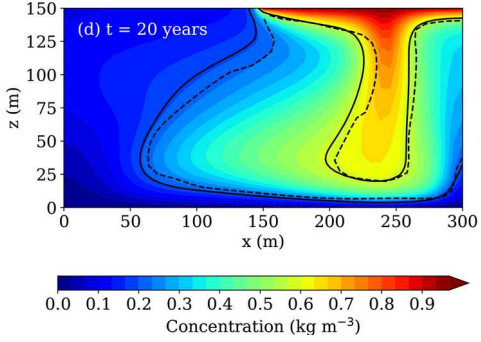
Adv. Geosci., 54, 67–77, 2020
https://doi.org/10.5194/adgeo-54-67-2020
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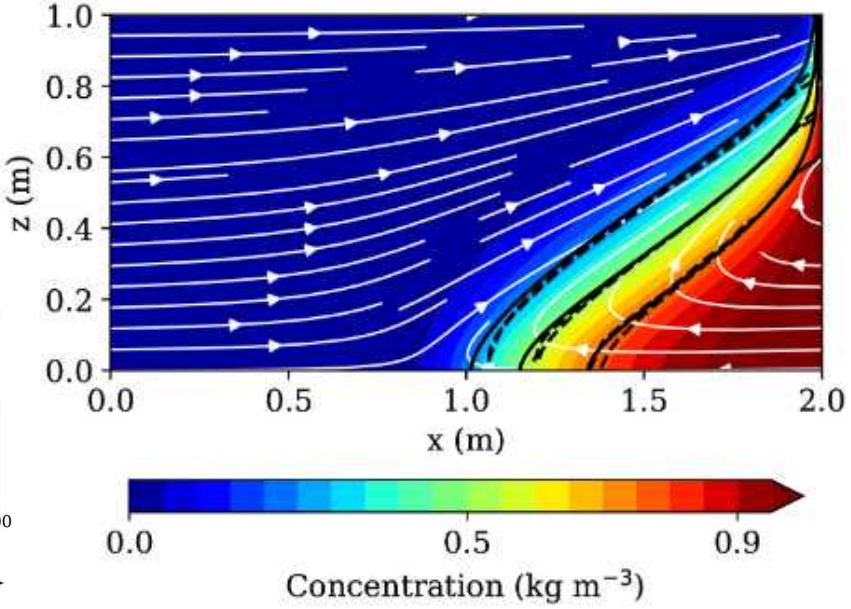
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Verification of a Python-based TRANsport Simulation Environment for density-driven fluid flow and coupled transport of heat and chemical species

Thomas Kempka[1,2]



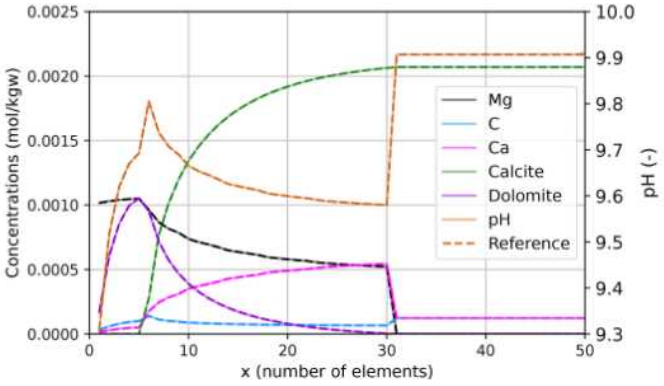




TRANSPORTSE-PHREEQC coupling verification

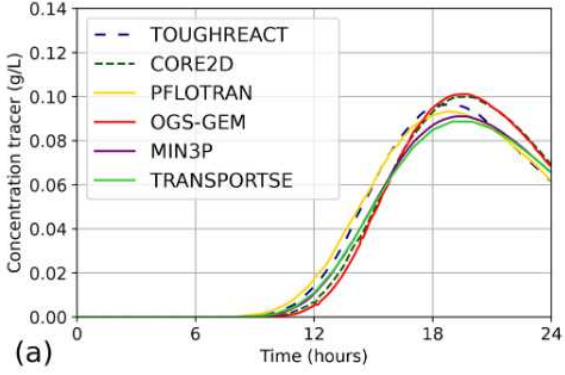
Adv. Geosci., 58, 19–29, 2022
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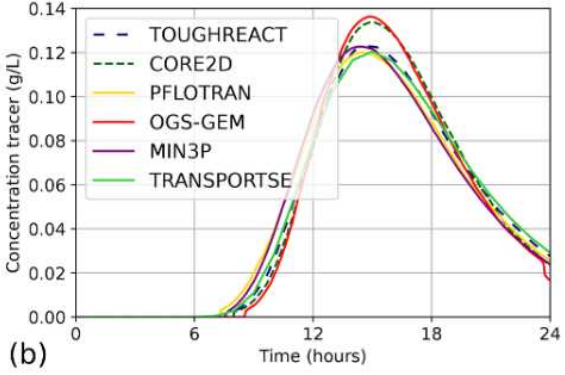
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Verification of TRANSPORT Simulation Environment coupling with PHREEQC for reactive transport modelling

Thomas Kempka[1,2], Svenja Steding[1], and Michael Kühn[1,2]





High density changes as well as solid salt dissolution and precipitation related to potash salt mining in Germany

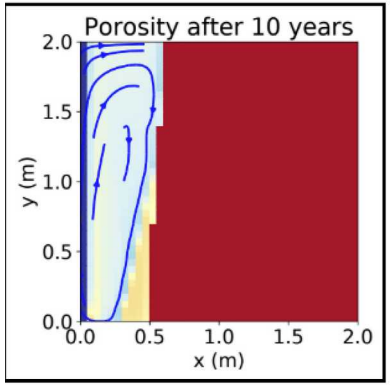
water

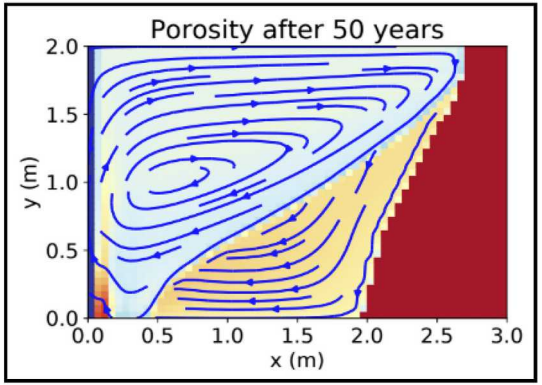
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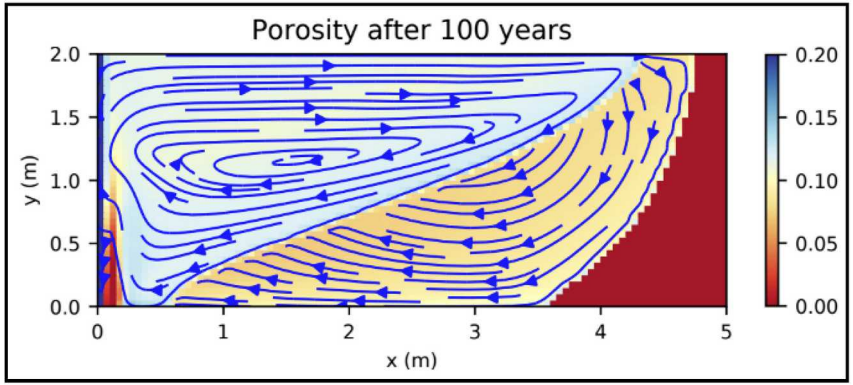
Article

Spatial and Temporal Evolution of Leaching Zones within Potash Seams Reproduced by Reactive Transport Simulations

Svenja Steding [1,2,*], Thomas Kempka [1,2], Axel Zirkler [3] and Michael Kühn [1,2]







Understanding the dynamics of salt dissolution in heterogeneous geological systems

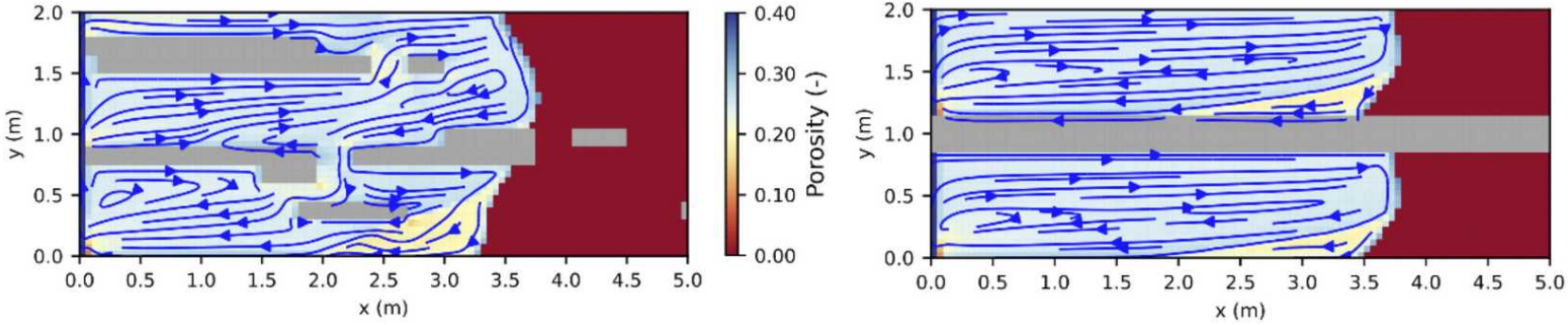
applied sciences

MDPI

Article

How Insoluble Inclusions and Intersecting Layers Affect the Leaching Process within Potash Seams

Svenja Steding [1,2,*], Thomas Kempka [1,2] and Michael Kühn [1,2]



Gas hydrate formation in laboratory-scale reactor

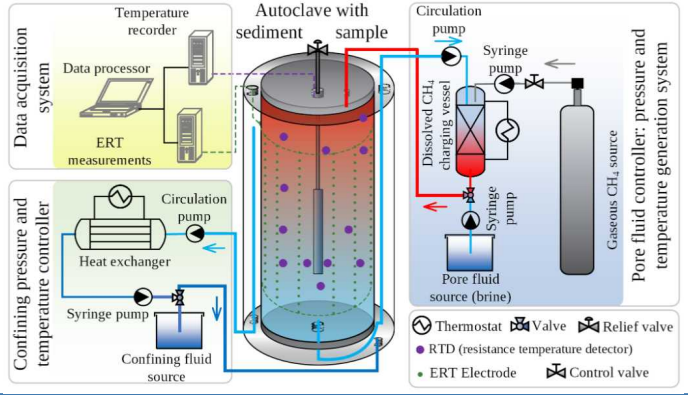
energies

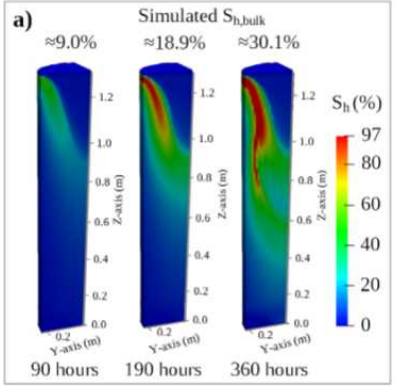
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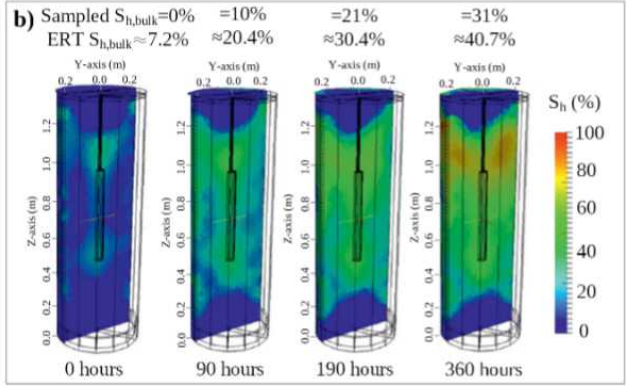
Article

Numerical Simulation of Hydrate Formation in the LArge-Scale Reservoir Simulator (LARS)

Zhen Li [1,2], Erik Spangenberg [1], Judith M. Schicks [1,3,*] and Thomas Kempka [1,2]







Permafrost freezing and thawing in view of methane gas hydrate formation in Mackenzie Delta

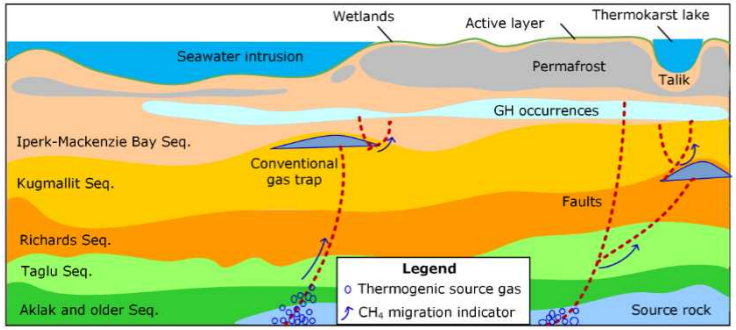
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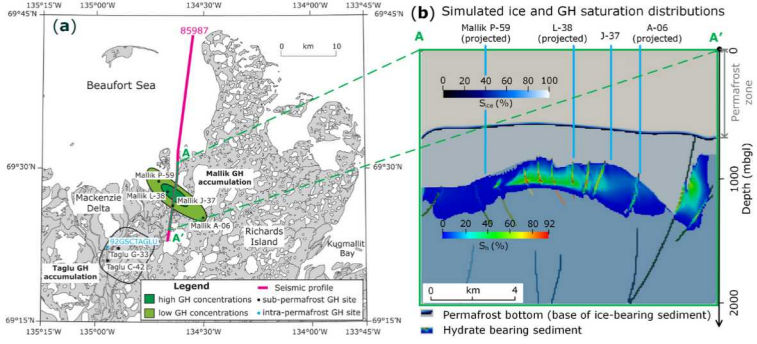
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Article

Numerical Simulation of Coastal Sub-Permafrost Gas Hydrate Formation in the Mackenzie Delta, Canadian Arctic

Zhen Li [1,2], Erik Spangenberg [1], Judith M. Schicks [1,3] and Thomas Kempka [1,2,*]





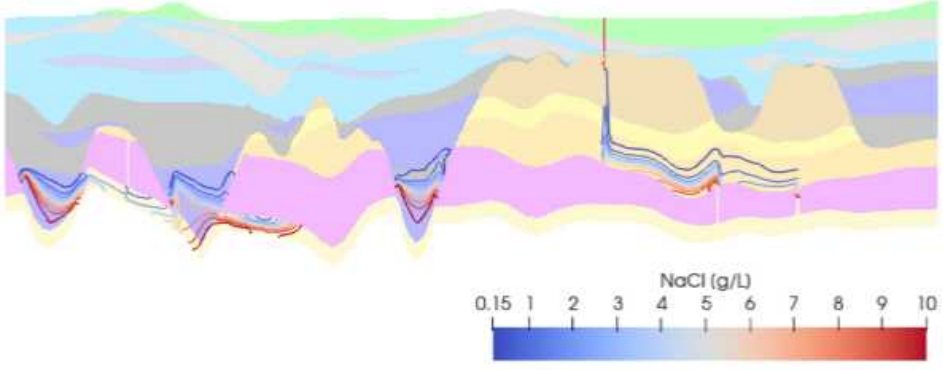
Groundwater salinisation resulting from freshwater production and climate change in Brandenburg

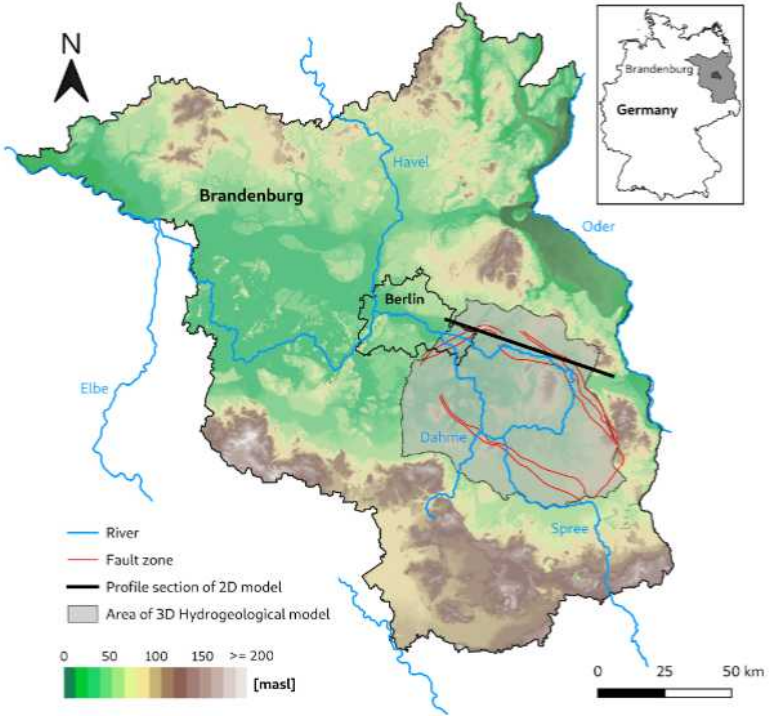
Adv. Geosci., 58, 47–54, 2022
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Upwelling mechanisms of deep saline waters via Quaternary erosion windows considering varying hydrogeological boundary conditions

Elena Chabab[1], Michael Kühn[1,2], and Thomas Kempka[1,2]





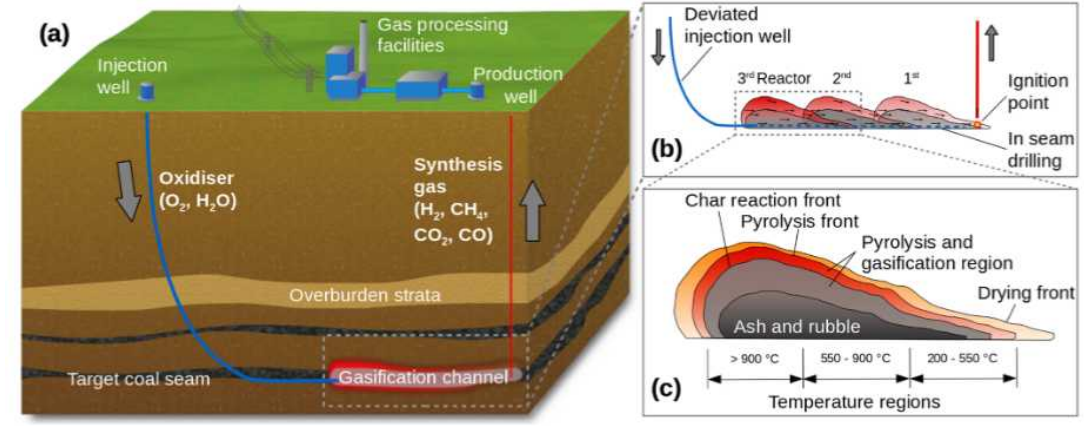
Groundwater contamination by subsurface utilisation

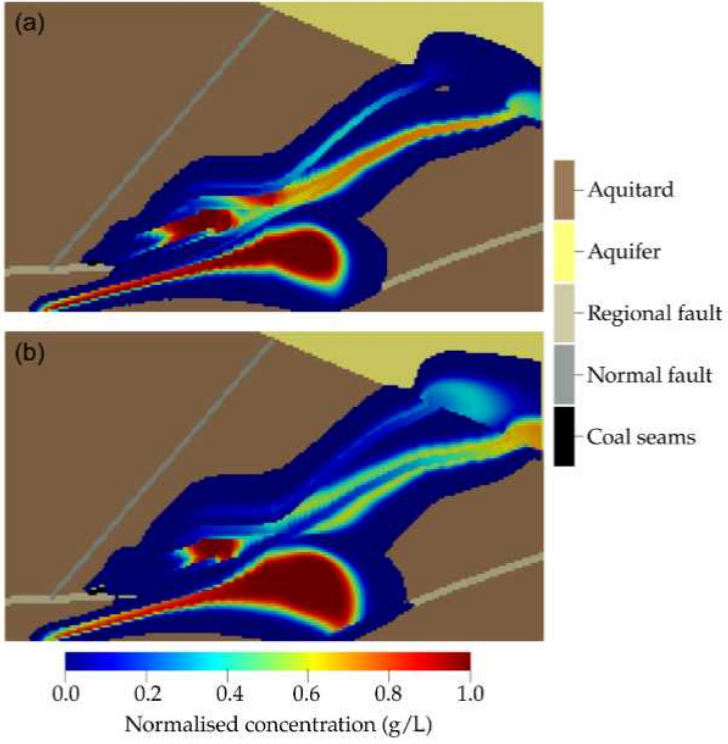
Adv. Geosci., 58, 55–66, 2022
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Numerical Analysis of Potential Contaminant Migration from Abandoned In Situ Coal Conversion Reactors

Christopher Otto[1], Svenja Steding[1], Morgan Tranter[1], Torsten Gorka[2], Mária Hámor-Vidó[3], Wioleta Basa[4], Krzysztof Kapusta[4], István Kalmár[5], and Thomas Kempka[1,6]



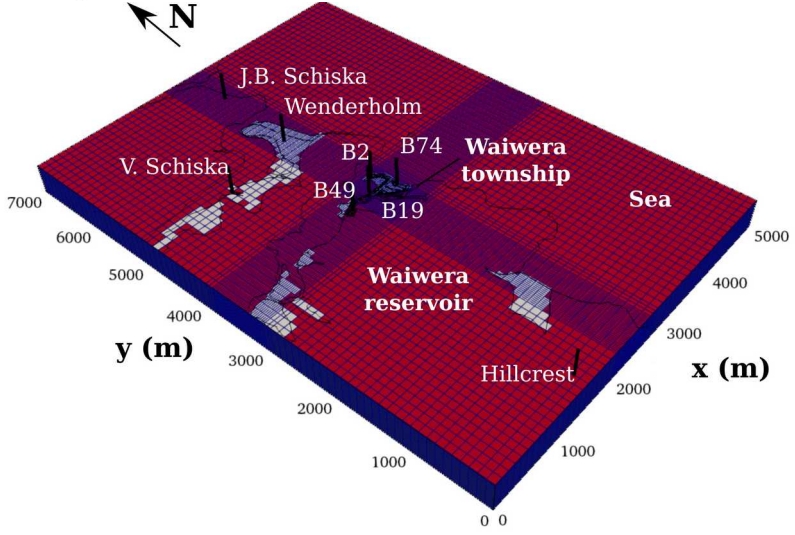


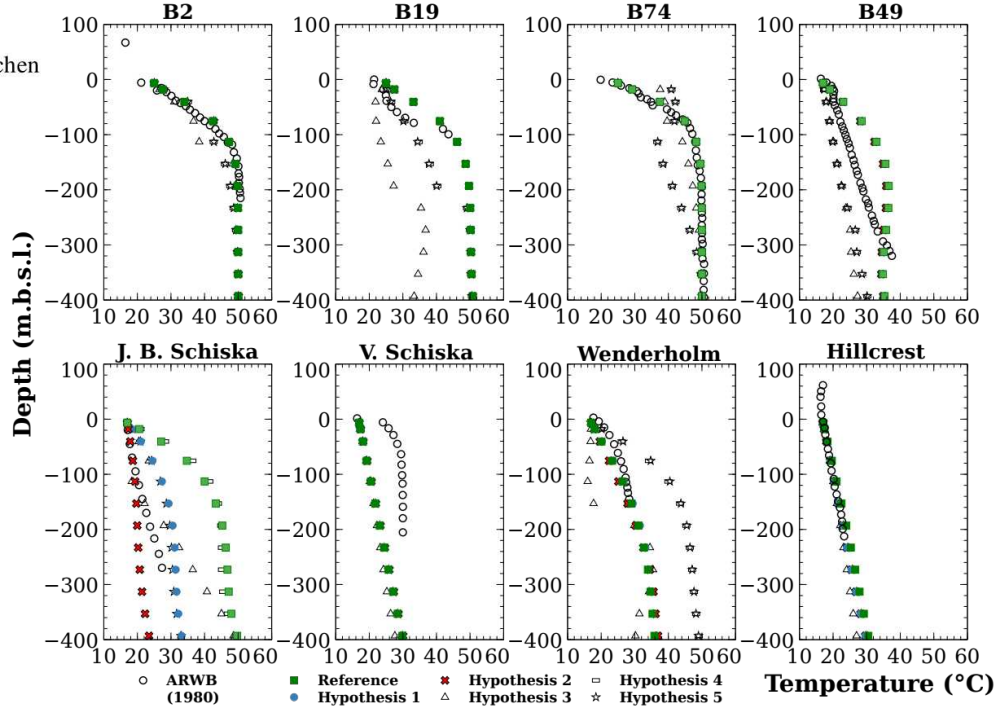
Geothermal evolution and costal groundwater salinisation

Title
Numerical simulation of spatial temperature and salinity distribution in the Waiwera geothermal reservoir, New Zealand

Titel
Numerische Simulation der räumlichen Temperatur- und Salinitätsverteilung im geothermischen Reservoir Waiwera, Neuseeland

Authors
Thomas Kempka[1,2*] and Michael Kühn[1,2]



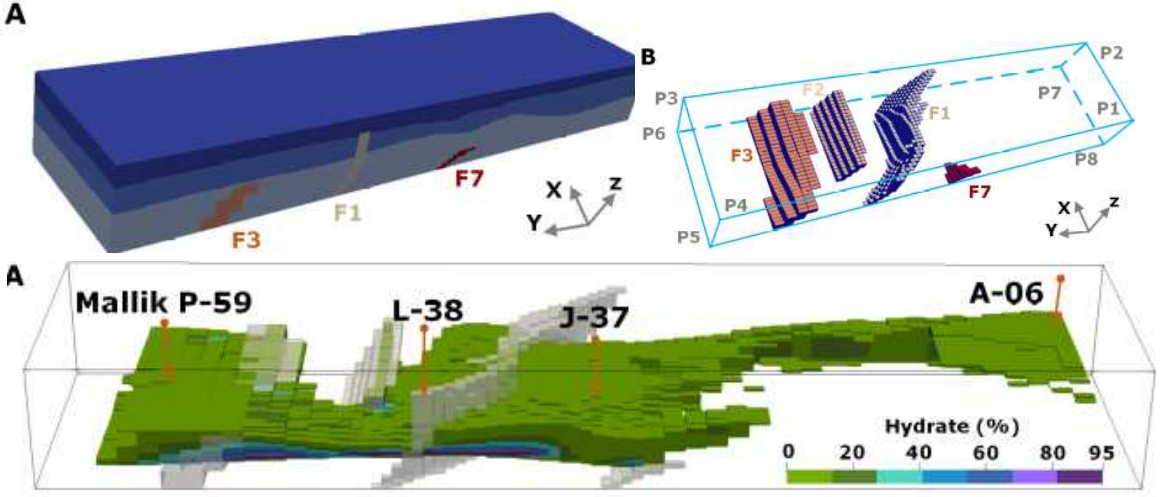


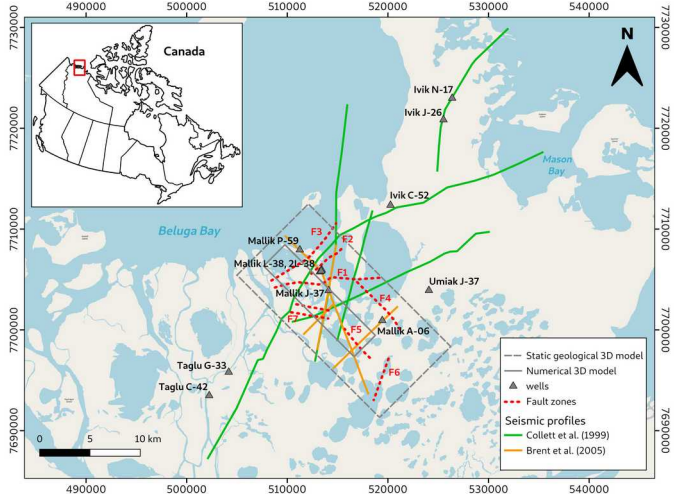
https://adgeo.copernicus.org/articles/58/31/2022/
Grundwasser (accepted for publication)

3D permafrost formation and thawing as well as hydrate deposit development at geological time scales

Geologic Controls on the Genesis of the Arctic Permafrost and Sub-Permafrost Methane Hydrate-bearing System in the Beaufort-Mackenzie Delta

Zhen Li [1,2], Elena Chabab [1], Erik Spangenberg [1], Judith M. Schicks [1,3] and Thomas Kempka [1,2*]





https://www.frontiersin.org/articles/10.3389/feart.2023.1148765/full